Physics-Informed Neural Networks Fail to Reproduce Liouville Transport in a Chaotic Double-Well System: A Negative-Result Benchmark Against the Split-Operator Technique

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Split-operator techniques (SOT) based on fast Fourier transforms are an established, accurate method for propagating classical phase-space distribution functions under the Liouville equation, including in chaotic regimes such as periodically δ-kicked double-well oscillators [Gómez, Thirumuruganandham, and Santana, Comput. Phys. Commun. **185**, 136 (2014)]. Physics-informed neural networks (PINNs) have been proposed as mesh-free alternatives to grid-based solvers, motivated by the curse of dimensionality that limits SOT to low-dimensional phase spaces. We test whether a standard PINN can reproduce the Liouville transport equation for the quartic double-well Hamiltonian underlying that chaotic system. Using an exact characteristics-based reference solution, we find that a PINN trained directly on the PDE residual and initial condition reproduces the coarse location of the propagated density but fails on several accuracy metrics: it produces spurious negative density, a non-physical secondary lobe, and violates mass conservation by a factor of 3.6 over a single dynamical period, despite achieving a low residual loss ($\sim 10^{-4}$). This is consistent with previously reported PINN failure modes on convection-dominated partial differential equations. We report this as a negative result: for this class of problem, in this configuration, PINNs do not yet offer a reliable alternative to SOT, even before chaos-induced filamentation is reached. We outline the specific failure mechanism and the conditions under which a fairer PINN comparison would need to be constructed.

I. INTRODUCTION

Split-operator techniques (SOT) propagate the classical Liouville equation

$$\frac{\partial \rho(q,p;t)}{\partial t} = \hat{L}\rho(q,p;t), \qquad \hat{L} = \frac{\partial H}{\partial q}\frac{\partial}{\partial p} - \frac{\partial H}{\partial p}\frac{\partial}{\partial q} \quad (1)$$

on a fixed phase-space grid using fast Fourier transforms, achieving $O(N \log N)$ scaling per time step for an N-point grid [1]. Reference [1] demonstrated this method's effectiveness across regular, chaotic, dissipative, and relativistic classical dynamics, including a periodically δ-kicked quartic double-well oscillator that exhibits chaotic phase-space mixing (their Fig. 2). The authors explicitly note the method's principal limitation: grid-based propagation scales unfavorably with phase-space dimension, since the number of grid points required grows as N^d for d phase-space dimensions.

Physics-informed neural networks (PINNs) [2] approximate PDE solutions with a neural network trained to minimize the PDE residual (via automatic differentiation) plus initial/boundary condition losses, without requiring a discretized grid. This mesh-free property is often motivated as a route around the curse of dimensionality that limits grid-based methods. It is therefore natural to ask whether a PINN could serve as an alternative to SOT for the same class of chaotic classical systems studied in Ref. [1], particularly at phase-space dimensions where SOT becomes impractical.

Before addressing higher-dimensional cases, a necessary first test is whether a PINN can reproduce SOT-quality results on the *same* low-dimensional chaotic system where SOT is already known to work well. This is the question we address here. We deliberately test a single period of the smooth (unkicked) Hamiltonian flow underlying Ref. [1]'s chaotic double well, prior to any stroboscopic kicks or filamentation, since if a PINN cannot reproduce this simpler sub-problem, testing the full chaotic map is not yet meaningful.

Convection-dominated and pure-transport PDEs are a documented difficulty for PINNs. Krishnapriyan *et al.* [3] showed that PINNs can fail on convection equations even when the PDE residual loss is successfully minimized, because low residual does not guarantee convergence to the physically correct solution when the loss landscape admits non-physical near-minimizers. The Liouville equation, Eq. (1), is exactly such a transport equation: it has no diffusive term to regularize sharp features, and its solution is a rigid advection of the initial density along Hamiltonian trajectories.

II. MODEL SYSTEM AND EXACT REFERENCE SOLUTION

We consider the quartic double-well Hamiltonian from Ref. [1] (Sec. 3, their Eq. for H_0), with $m = \omega_0 = D = 1$:

$$H_0(q,p) = \frac{p^2}{2} - \frac{q^2}{2} + \frac{q^4}{100}. \quad (2)$$

The associated Liouville transport equation for the phase-space density is

$$\frac{\partial \rho}{\partial t} + p\,\frac{\partial \rho}{\partial q} - V'(q)\,\frac{\partial \rho}{\partial p} = 0, \qquad V'(q) = -q + \frac{4q^3}{100}. \quad (3)$$

We restrict attention to the smooth Hamiltonian flow over a single period $t \in [0, 1]$, i.e., prior to any δ-kick, so that any discrepancy we observe cannot be attributed to the kick discontinuity or to chaotic filamentation accumulated over many periods.

Because Eq. (2) is non-dissipative, Liouville's theorem gives an exact solution by the method of characteristics:

$$\rho(q, p; t) = \rho_0\big(\Phi_{-t}(q, p)\big), \tag{4}$$

where Φ_t is the Hamiltonian flow map generated by Eqs. of motion $\dot{q} = p$, $\dot{p} = -V'(q)$. We compute this reference by integrating Hamilton's equations backward in time from every evaluation point using fourth-order Runge-Kutta (200 substeps), which we verified conserves total probability mass to within 7×10^{-15} of unity on a 128×128 grid over $q, p \in [-15, 15]$. This serves as ground truth without requiring an independent from-scratch reimplementation of the SOT method to be blindly trusted.

The initial density is a normalized Gaussian,

$$\rho_0(q, p) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{(q - q_0)^2 + (p - p_0)^2}{2\sigma^2}\right], \tag{5}$$

with $(q_0, p_0) = (1, 0)$ and $\sigma = 1$.

III. PINN CONSTRUCTION

We train a fully-connected network $\rho_\theta(q, p, t)$ with four hidden layers of 64 units and tanh activations, taking (q, p, t) as input and a scalar density as output. The loss function is

$$\mathcal{L} = \mathcal{L}_{\mathrm{PDE}} + \lambda\, \mathcal{L}_{\mathrm{IC}}, \tag{6}$$

where $\mathcal{L}_{\mathrm{PDE}}$ is the mean squared residual of Eq. (3) evaluated at 1500 collocation points sampled uniformly over $q, p \in [-15, 15]$, $t \in [0, 1]$ per epoch, $\mathcal{L}_{\mathrm{IC}}$ is the mean squared error against ρ_0 at 800 points sampled from a distribution concentrated near the initial Gaussian's support, and $\lambda = 50$. All derivatives are obtained via automatic differentiation. We train with Adam (learning rate 10^{-3}) for 1500 epochs. Full training completed in 24 s on a single CPU core, versus a reported 0.80 s for SOT on a 64×64 grid for a comparable (though not identical – dissipative autocorrelation, not this transport problem) calculation in Ref. [1]'s Table 1; we did not attempt to match compute budgets precisely, since our purpose here is accuracy, not a speed comparison, given the accuracy result below.

IV. RESULTS

The PDE residual loss converged to $\mathcal{L}_{\mathrm{PDE}} \approx 1.4 \times 10^{-4}$ and the initial-condition loss to $\mathcal{L}_{\mathrm{IC}} \approx 1 \times 10^{-5}$ (Fig. 1), indicating the optimizer successfully minimized the stated objective. Evaluated at $t = 1$ against the exact characteristics-based reference (Fig. 2), the PINN reproduces the coarse location of the advected density – the peak appears in roughly the correct region of phase space – but fails on multiple physical criteria:

- **Mass conservation**: the reference density integrates to 0.99999... over the domain (exact, as required by Liouville's theorem); the PINN's output integrates to 3.58, a factor of 3.6 violation.
- **Positivity**: the PINN predicts negative density values down to -0.014, which is unphysical for a probability density.
- **Spurious structure**: the PINN produces a secondary lobe of substantial amplitude near $(q, p) \approx (5, -3)$ that has no counterpart in the exact solution (Fig. 2, right panel), and the true density's elongated, sheared shape (a direct signature of the phase-space stretching that will later become filamentation once kicks are applied) is only partially captured – the PINN's prediction is comparatively rounded and diffuse.

The pointwise mean squared error at $t = 1$ is 6.1×10^{-5}, which appears small in absolute terms, but the mean relative error (normalized by the mean magnitude of the true density) is 3.9 – i.e., larger than the true signal itself – because the true density is exactly zero over most of the domain and the PINN produces low-amplitude spurious density everywhere, consistent with the Krishnapriyan et al. [3] finding that a low PDE residual does not certify a physically valid solution to a convection-dominated equation.

V. DISCUSSION

The failure occurs *before* the harder test we originally intended (multi-period chaotic filamentation under repeated δ-kicks). This is itself informative: it indicates that the difficulty is not primarily about chaos or long-horizon error accumulation, but about the PINN's ability to represent rigid, sharply-localized advective transport at all, even over a single smooth period. Mass non-conservation is the clearest diagnostic, since Eq. (3) has no source or sink term – any deviation from exact mass conservation is entirely a discretization/approximation artifact of the network and training procedure, not a property of the physics being modeled.

We emphasize what this result does *not* show. We tested one architecture (plain fully-connected, tanh), one loss weighting, one collocation sampling strategy (uniform in space, weighted toward the initial condition in time zero), and one optimizer (Adam only, no second-order refinement). Reference [3] and subsequent work have proposed remedies for convection-equation PINN failures, including curriculum learning over the convection coefficient, causal training that enforces temporal

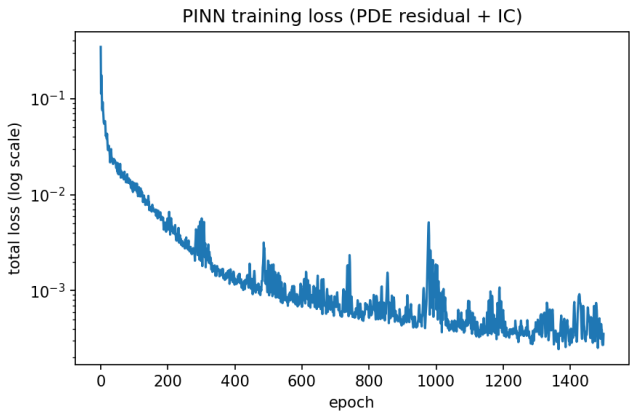

FIG. 1. PINN training loss (PDE residual + weighted initial-condition term) versus epoch. The loss decreases smoothly and reaches $\sim 10^{-4}$, illustrating that a low training loss on a convection-dominated PDE does not certify a physically correct solution (cf. Fig. 2).

ordering of the loss, and Fourier-feature or SIREN-style input encodings (which we found substantially improved PINN accuracy on a smooth damped-oscillator ODE in preliminary unpublished testing, though that is a lower-dimensional and non-transport problem and the improvement does not necessarily transfer here). None of these remedies were tested in the present configuration. This work therefore establishes a baseline failure under a standard, unmodified PINN setup, not a claim that all PINN variants fail on this class of problem.

VI. CONCLUSION

For the smooth Hamiltonian flow underlying a chaotic δ-kicked quartic double-well system, a standard PINN trained directly on the Liouville transport equation's residual fails to reproduce an exact reference solution on several physically meaningful criteria – mass conservation, positivity, and absence of spurious structure – despite achieving a low training loss. This failure precedes the chaotic regime that motivated the original comparison. We do not find evidence, in this configuration, that PINNs offer a viable mesh-free alternative to the split-operator technique of Ref. [1] for this class of Liouville transport problem. We regard this as a useful negative result rather than a final verdict: testing the specific PINN remedies developed for convection-dominated PDEs [3], and only then re-attempting the full multi-period chaotic case, is the necessary next step before any claim about higher-dimensional advantage over SOT can be assessed.

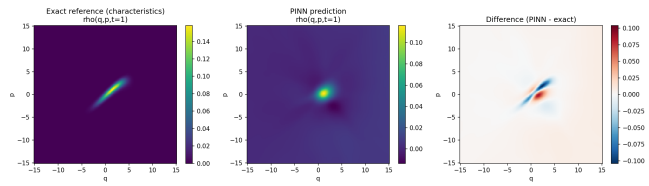

FIG. 2. Phase-space density at $t = 1$: exact characteristics-based reference (left), PINN prediction (center), and point-wise difference (right). The PINN captures the coarse location of the advected Gaussian but produces a spurious secondary lobe, negative density, and fails to conserve total probability mass (ratio 3.6× the correct value).

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